Introduction

The COVID-19 pandemic caused unprecedented disruptions to public transportation systems across the United States. In New York City, subway service was significantly altered beginning in March 2020, with reduced schedules, overnight closures, and changes in ridership patterns. Although full 24/7 subway service eventually resumed, it is not immediately clear whether system reliability returned to pre-pandemic levels or followed a different trajectory during the recovery period.

Defining the COVID and Recovery Periods

To ensure my analysis reflected real-world operational changes, I defined three periods using specific policy milestones:

- **Pre-COVID period (before March 20, 2020)**
  March 20, 2020 marks the day former Governor Andrew Cuomo ordered the closure of all non-essential businesses in New York State. Although the subway remained open, service levels were significantly reduced and overnight shutdowns were introduced.
- **COVID period (March 20, 2020 to May 17, 2021)**
  This period represents the height of pandemic-era service disruptions, reduced ridership, and modified operations implemented for health and safety reasons.
- **Recovery period (after May 17, 2021)**
  On May 17, 2021, the MTA resumed **24/7 subway service** for the first time since the start of the pandemic. I treat this date as the beginning of the system's operational recovery.

Data Cleaning and Preparation

The raw dataset contained **2,856 observations** across **12 variables**, including ridership levels, delay measures, and categorical fields such as subway line and division. While there were no missing values in numerical columns, the subway line labels were inconsistent, containing a mix of numbers, letters, spaces, and special cases.

To address this:

- I standardized all subway line labels by removing spaces, converting text to uppercase, and mapping known shuttle variants to consistent names.

- I created a new categorical variable called **`line_type`**, grouping lines into four categories:
    - **NUMBER** (e.g., 1–7)
    - **LETTER** (e.g., A, C, Q)
    - **COMBINED** (e.g., J/Z)
    - **SHUTTLE** (e.g., Rockaway Shuttle)
- For modeling purposes, I create a variable called **month_index**, defined as the number of months since May 17, 2021 (the start of the recovery period). This allows me to measure whether performance trends upward or downward over time, independent of calendar year labels.

This step allowed me to compare performance at a higher structural level rather than focusing on individual lines.

After cleaning and feature engineering, I saved the processed dataset as `mta_cleaned.csv` and used it for all subsequent analysis.

Overall Reliability Trends Over Time

When I plotted average customer journey time performance by month, a clear pattern emerged.

Before COVID, average performance hovered around **0.86**, indicating relatively stable service. During the COVID period, average performance actually **increased to approximately 0.87**, which may seem surprising at first. However, this increase coincided with historically low ridership levels, meaning trains faced fewer delays caused by congestion.

During the recovery period, average performance **declined to about 0.85**, a drop of roughly **0.02 points** compared to the COVID period. While this change may appear small, in a metric bounded between 0 and 1, a two-point drop represents a meaningful reduction in system-wide reliability.

Phase-Level Performance Differences

Looking at summary statistics by phase provided further insight:

- During the **pre-COVID period**, the mean performance was **0.861**, with a standard deviation of **0.053**, indicating moderate variability.
- During **COVID**, the mean increased to **0.866**, while variability remained nearly unchanged.

- During the **recovery period**, the mean dropped to **0.850**, and the standard deviation increased to **0.062**, making it the most volatile period in the dataset.

This tells me that not only did performance decline during recovery, but it also became **less consistent month to month**.

Peak vs. Off-Peak Performance

Service period played a major role in performance outcomes.

Across all phases, **peak service consistently outperformed off-peak service**:

- During COVID, peak performance averaged **0.877**, while off-peak averaged **0.855**, a difference of **0.022**.
- During recovery, peak performance declined to **0.862**, while off-peak dropped further to **0.838**, widening the gap.

This suggests that the system prioritized peak-hour reliability as ridership returned, while off-peak service recovered more slowly.

Performance by Line Type

Breaking performance down by line type revealed substantial structural differences:

- **Shuttle lines** consistently performed the best, with recovery-period performance near **0.95**, only slightly below their pre-COVID levels.
- **Numbered lines** performed relatively well, averaging **0.88** pre-COVID and **0.88** during recovery.
- **Letter and combined lines** experienced the largest declines, with recovery averages closer to **0.82**, down from pre-COVID values near **0.83–0.84**.

These results indicate that recovery has not been uniform and that certain types of service face persistent reliability challenges.

Regression Modeling: COVID vs. Recovery

To better understand *what drives performance*, I built separate linear regression models for the **COVID** and **recovery periods**.

COVID Model Results

The COVID-period regression achieved an **R^2 of approximately 0.72**, meaning the model explained about **72% of the variation** in performance during that period. This relatively high value suggests that system behavior during COVID was more predictable, likely due to simplified service patterns and reduced ridership.

Key findings from the COVID model:

- **Peak service** improved performance by roughly **0.03** compared to off-peak.
- **Shuttle lines** showed a very large positive effect, improving performance by more than **0.10** relative to baseline lines.
- **Passenger volume had a weaker negative effect**, reflecting the unusually low ridership environment.

Recovery Model Results

The recovery-period model had a slightly lower **R^2 of about 0.66**, meaning it explained **66% of performance variation**. This drop indicates that subway operations became more complex and less predictable as service normalized.

Key recovery model insights:

- Shuttle lines still had a strong positive effect, improving performance by roughly **0.115**, confirming their continued reliability.
- Peak service improved performance by about **0.026**, similar to COVID but slightly weaker.
- Passenger volume had a **more negative effect than during COVID**, indicating that increased ridership now actively constrained reliability.
- B Division lines performed approximately **0.046 points worse** than A Division lines, highlighting structural differences in recovery outcomes.

Comparing COVID and Recovery Models

Comparing the two models reveals an important shift:

- During COVID, performance was **higher and more predictable**, driven largely by reduced demand.
- During recovery, performance became **lower and more variable**, with ridership levels playing a stronger role.

- Structural features such as line type and division became more important during recovery, suggesting that operational challenges resurfaced as the system returned to full service.

Line-by-Line Recovery Patterns (Trend vs. Level)

There are **two different "ways" a line can look good or bad** in recovery:

1. **Recovery trend (slope):** *Is the line improving month-to-month during recovery?*
2. **Recovery level (gap vs. pre-COVID):** *Has the line returned to (or surpassed) its pre-pandemic baseline on average?*

These are related, but they are **not the same thing**, and it's possible for a line to look strong on one metric while looking weak on the other. That's why I'm careful to separate the findings into "best recovery trends" and "best recovery levels."

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1) Best Recovery Trends: Lines improving the fastest during recovery

Based on the **recovery slope per line**, the lines that recovered the best *in terms of momentum* were **E, C, W, A, and M**. In my output, these lines had the **highest positive slopes** (i.e., the largest values for `slope_per_month`).

- A **positive slope** means the line's average customer journey performance was **increasing over time** during the recovery period.
- A **larger slope** means the line's reliability was improving **faster month-to-month** compared to other lines.

For example, the **E line** stands out because its `slope_per_month` was about **0.00193**, and its net change from the start to end of the recovery period was about **+0.114**. That's a sizable improvement in this performance scale. The **C line** also had a strong positive slope (about **0.00118**), showing that it was trending upward steadily across the recovery window.

Important nuance:
When I say "recovered the best," I'm specifically talking about **the trend** (the *rate of improvement*), not necessarily the highest overall performance level. A line can improve quickly but still not fully catch up to where it used to be (this becomes important for interpreting the C line later).

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2) Worst Recovery Trends: Lines that stagnated or got worse during recovery

The lines that recovered the worst (meaning they did not improve, or actually declined over time in recovery) were **JZ, 6, 2, S_FKLN, and L**, because their recovery slopes were **negative**.

What a **negative slope** means:

- A negative `slope_per_month` means that, month-by-month, the line's average performance was **trending downward** during recovery.
- In other words, even though the subway system was "back" to 24/7 operations, these lines were **getting less reliable over time** (or at best, not sustaining early recovery gains).

Clarification:
A negative recovery slope does not automatically mean a line was "bad" the entire recovery period. It could mean:

- the line started recovery at a decent level, then gradually declined, or
- the line improved early and later regressed, pulling the trend negative overall.

This is why I treat slope as a **direction + momentum** measure, not an absolute score.

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3) Lines that exceeded pre-pandemic performance: "Recovered above baseline"

Next, I used the **pre-COVID vs. recovery gap** table to measure whether lines have, on average, returned to their baseline. Here, the key column is the gap:

```
gap_recovery_minus_pre = recovery_mean - pre_covid_mean
```

So:

- A **positive gap** means the line's average performance during recovery was **higher than pre-COVID**.
- A **negative gap** means it was still **below pre-COVID** (i.e., recovery is incomplete relative to its own baseline).

Based on the results, the lines that were **doing better in recovery than pre-pandemic** were **L, S_42ND, S_FKLN, 5, and 4**, because their recovery gaps were **positive**.

For instance:

- The **L line** showed a positive gap (recovery above pre-COVID), which suggests that its recovery period average reliability was **better than its pre-pandemic average**.
- **S_42ND** and **S_FKLN** also posted positive gaps, which is notable because shuttles are often assumed to be more fragile operationally—yet here they appear to have maintained or exceeded baseline performance on average.

Important nuance:
A line can have a positive gap **even if its recovery slope is negative**. That means:

> The line may have recovered quickly and reached a strong level, but then plateaued or drifted downward later.

This is exactly the kind of situation where trend (slope) and level (gap) tell **different but complementary stories**.

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4) Lines furthest below pre-pandemic baseline: "Incomplete recovery"

Finally, the lines performing worst post-pandemic (in the sense of being the most below their pre-COVID baseline) were **F, C, R, S_ROCK, and D**, because they had the most negative recovery gaps.

What that means:

- These lines remain the **furthest below their own pre-pandemic reliability levels**, even years into the recovery period.
- This suggests the recovery experience was uneven—some lines returned to baseline or surpassed it, while others remain persistently worse than before COVID.

Key nuance: The C line looks "mixed" in a meaningful way
The **C line** shows up in two places:

- It is one of the **best recovery trends** (strong positive slope), meaning it is improving steadily.
- But it is also among the **worst recovery gaps**, meaning it is still below its pre-COVID average.

This is not a contradiction. It means:

> The C line is improving over time, but it started recovery from a low point and has not fully caught up to its pre-pandemic baseline yet.

This "high slope + negative gap" pattern is analytically important because it implies:

- Progress is happening (good trend),
- but the line still has a "distance to go" (bad level).

In contrast, a line like **L** can show the opposite pattern:

- Positive gap (better than baseline),but a weaker or negative slope (not improving anymore, possibly plateauing).

That combination tells a different recovery story:

> The line recovered early and is now stable (or slightly declining), but still above its old baseline overall.

Summary: What I can confidently conclude from the line-level results

Putting these findings together, I conclude that recovery is **not one-dimensional**. The line-level analysis shows at least three distinct recovery "types":

1. **Strong improvement momentum (largest positive slope):** lines like **E, C, W, A, M** are trending upward fastest.
2. **Recovery regression (largest negative slope):** lines like **JZ, 6, 2, S_FKLN, L** are trending downward during recovery.
3. **Recovery level vs. baseline (gap):** some lines have **fully recovered (or exceeded)** pre-pandemic performance (e.g., **L, S_42ND, S_FKLN, 5, 4**), while others remain **persistently below baseline** (e.g., **F, C, R, S_ROCK, D**).

Most importantly, I avoid treating "recovery" as a single ranking. A line can be:

- improving quickly but still below baseline (C), or
- above baseline but no longer improving (L), or
- trending downward while also below baseline (worst-case scenario).

This interpretation helps me describe subway recovery in a way that is both **accurate and nuanced**, rather than oversimplifying the story into "best" and "worst" lines without context.

Conclusion

> Overall, my analysis shows that NYC subway reliability did not follow a simple decline-and-rebound pattern. Instead, performance *peaked during the pandemic*, largely due to suppressed demand and reduced service complexity, and then *declined during the recovery period* as ridership returned and full operations resumed.
>
> By analyzing monthly trends, phase-level averages, and line-specific recovery slopes, I found that although the system has largely stabilized since mid-2021, recovery remains uneven across the network. Off-peak service, lettered lines, and certain divisions continue to lag behind their pre-pandemic baselines, while other lines show strong upward momentum despite remaining below historical performance levels.

Importantly, this uneven recovery is not captured by ridership totals alone. My results show that some lines with increasing ridership are still experiencing declining or stagnant reliability, while others have exceeded pre-pandemic performance but show limited continued improvement. This distinction highlights the need to evaluate recovery using both **trend-based metrics** (how performance is changing over time) and **level-based metrics** (how current performance compares to historical baselines).